



Artificial Intelligence in Medicine:

Foundations and Clinical Applications

Instructor



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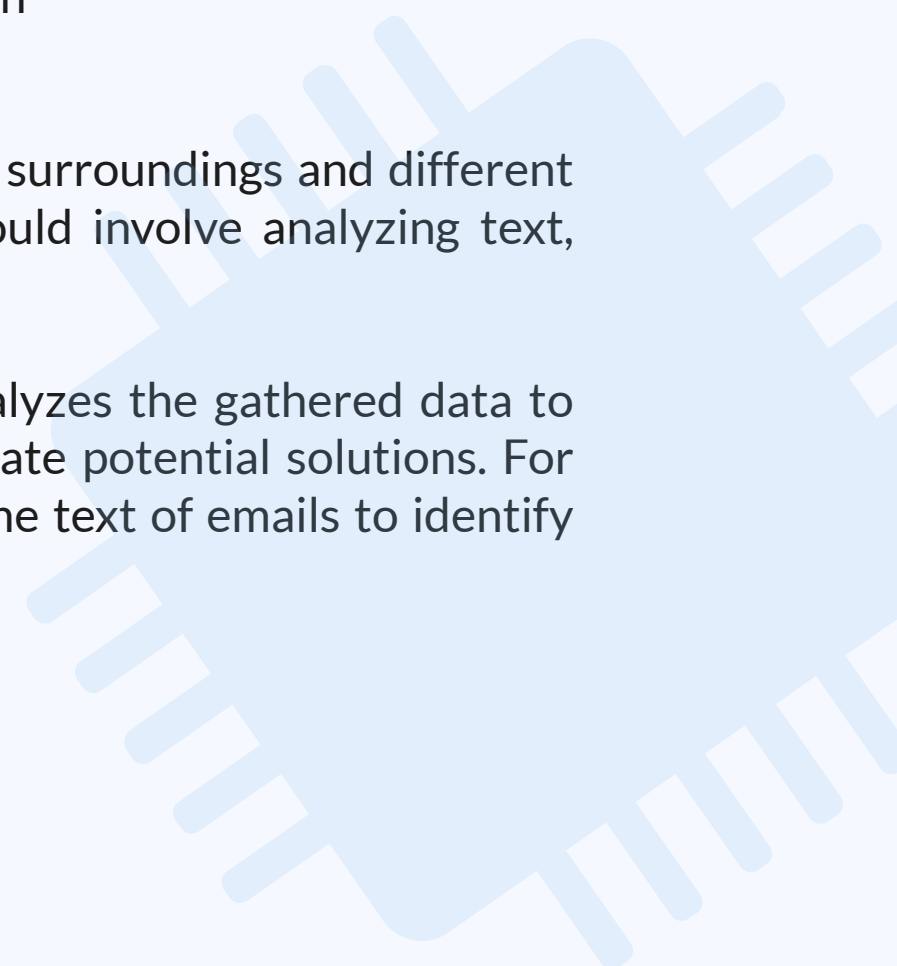
What is agentic AI?

❑ **Agentic AI** is an advanced form of artificial intelligence focused on autonomous decision-making and action. Unlike traditional AI, which primarily responds to commands or analyzes data, agentic AI can set goals, plan, and execute tasks with minimal human intervention

❑ Key concepts of agentic AI

❑ **Perception:** Agentic AI starts by gathering information from its surroundings and different sources, such as sensors, databases, and user interfaces. This could involve analyzing text, images, or other forms of data to understand the situation.

❑ **Reasoning:** Using a large language model (LLM), agentic AI analyzes the gathered data to understand the context, identify relevant information, and formulate potential solutions. For example, if the goal is to schedule a meeting, the LLM can parse the text of emails to identify attendees, available times, and the meeting's purpose.



What is agentic AI?

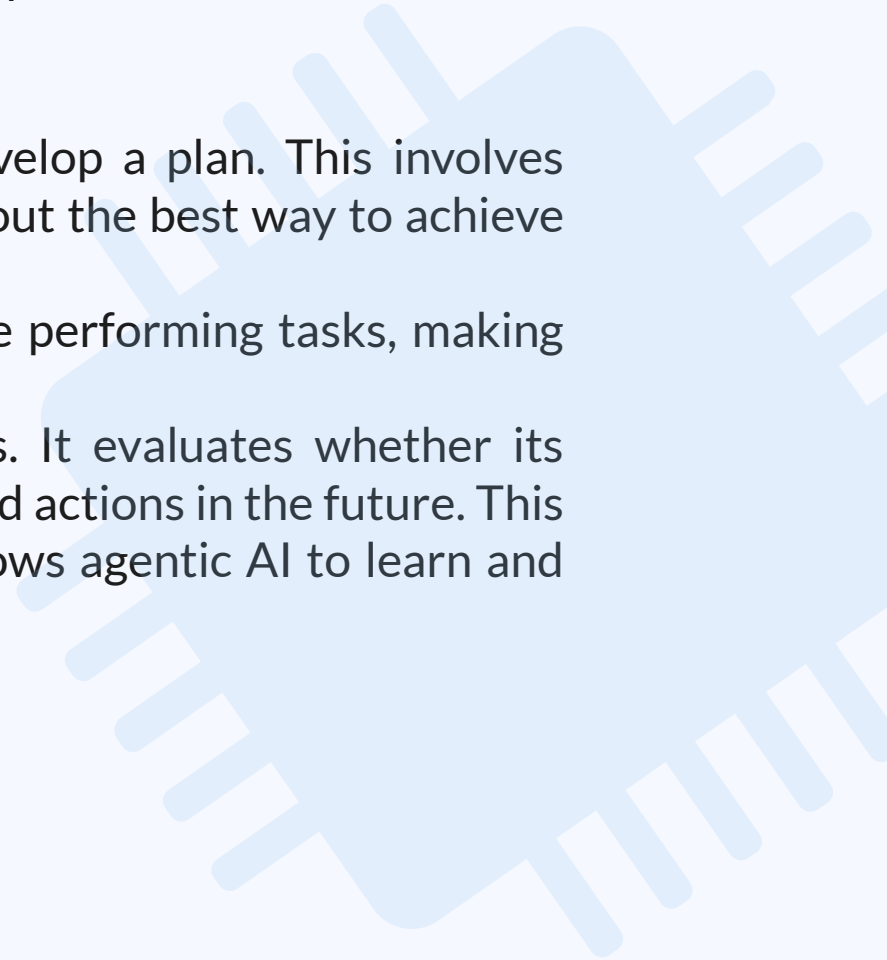
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❑ Key concepts of agentic AI

❑ **Planning:** The AI then uses the information it gathered to develop a plan. This involves setting goals, breaking them down into smaller steps, and figuring out the best way to achieve them.

❑ **Action:** Based on its plan, the AI takes action. This could involve performing tasks, making decisions, or interacting with other systems.

❑ **Reflection:** After taking action, the AI learns from the results. It evaluates whether its actions were successful and uses this feedback to adjust its plans and actions in the future. This continuous cycle of perception, planning, action, and reflection allows agentic AI to learn and improve over time



What is agentic AI?

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The role of agentic artificial intelligence in healthcare: a scoping review

[Bernardo G. Collaco](#), [Syed Ali Haider](#), [Srinivasagam Prabha](#), [Cesar A. Gomez-Cabello](#), [Ariana Genovese](#), [Nadia G. Wood](#), [Sanjay P. Bagaria](#), [Narayanan Gopala](#), [Cui Tao](#) & [Antonio Jorge Forte](#) 

[npj Digital Medicine](#) **9**, Article number: 345 (2026) | [Cite this article](#)

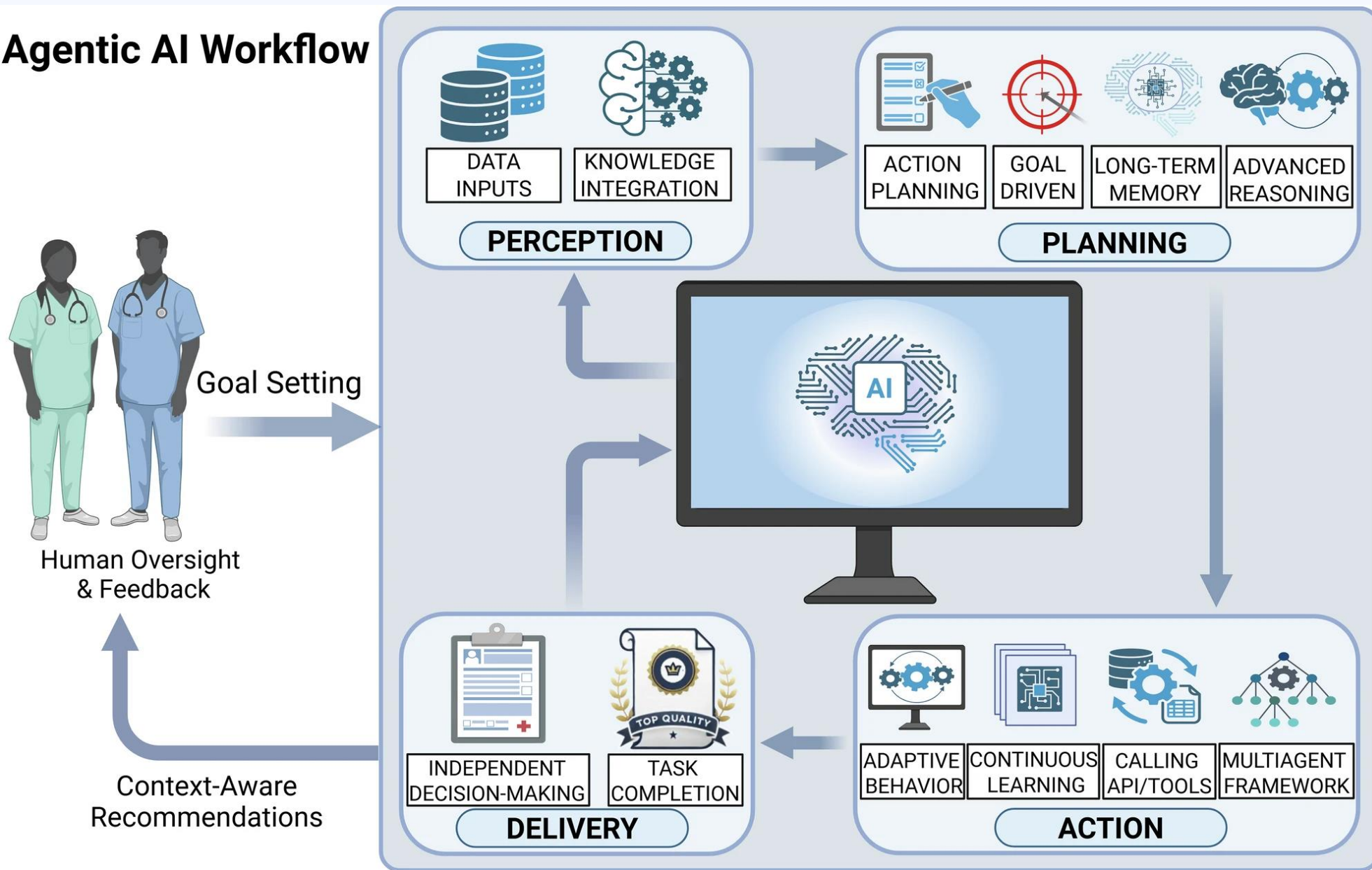
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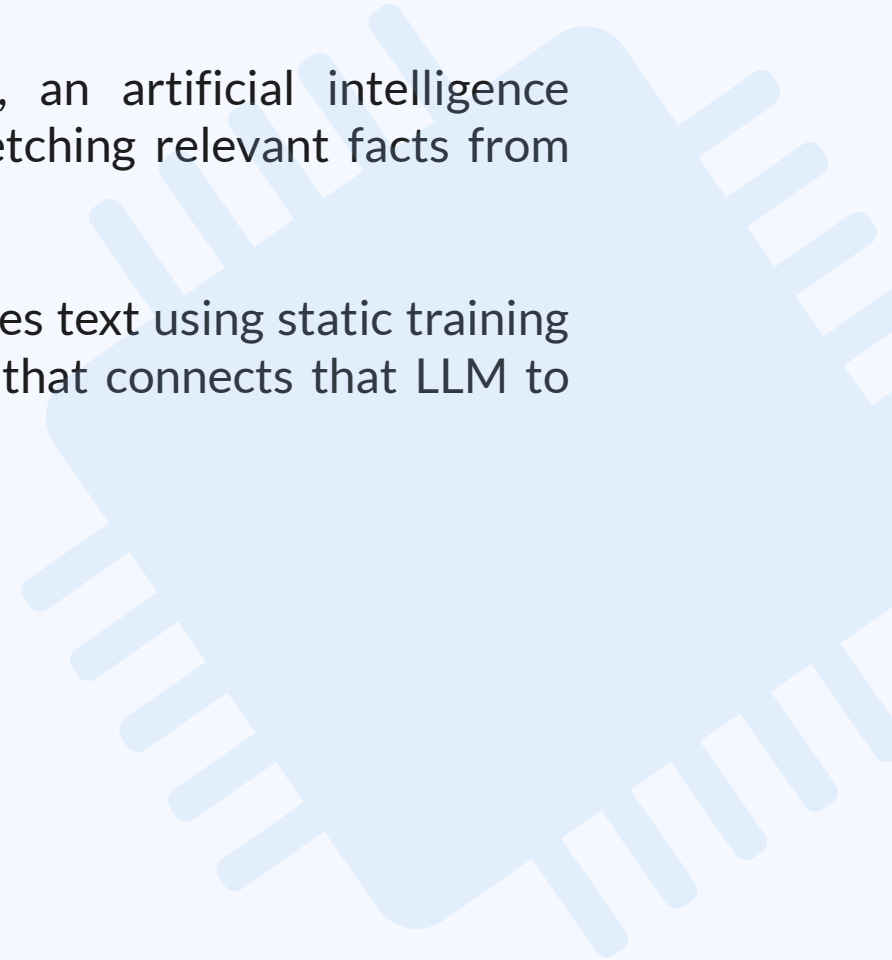


Agentic AI Workflow



What is RAG? (Medical Analogy)

- ❑ **Standard LLM:** An open-book exam where the AI relies only on what it memorized during training (which might be outdated or hallucinated).
- ❑ **RAG System:** RAG stands for Retrieval-Augmented Generation, an artificial intelligence technique that improves large language model (LLM) responses by fetching relevant facts from external data sources before generating an answer.
- ❑ A Large Language Model (LLM) is a standalone AI brain that generates text using static training data, whereas Retrieval-Augmented Generation (RAG) is a technique that connects that LLM to external data sources to fetch real-time facts before answering.



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MA

Mr. Angelo Little

Today at 11:09 AM

Generate a sales email for <https://zapier.com/>.

Automated Sales Lead Emails

Today at 11:10 AM

Sure! I'll lookup that company and generate a sales email.

Automated Sales Lead Emails

 Agents



PK

Parry Klein

Today at 11:09 AM

Whenever a new intake call is scheduled, prep a doc and share it with the sales team in Slack.

Biz Dev Call Briefer

Today at 11:10 AM

Absolutely! I'll generate a call prep doc, post it in #bizdev, and log the meeting in your tracking sheet every time a new intake meeting gets scheduled.

Biz Dev Call Briefer

 Agents



YD

Yvonne Okuneva

Today at 11:09 AM

Can you check the new Google Doc I created for brand compliance?

Brand Guidelines Agent

Today at 11:10 AM

Sure! I'll review the document for adherence to our brand guidelines and send you feedback shortly.

Brand Guidelines Agent

 Agents



CH

Cathy Hill-Sawyn

Today at 11:09 AM

Please analyze these texts for our brand voice.

Brand Voice Analyzer

Today at 11:10 AM

Sure! I'll analyze the texts you provided and create a comprehensive brand voice style guide along with a summary of the key findings.

Brand Voice Analyzer

 Agents



MD

Ms. Devitt Demsek

Today at 11:09 AM

Write a follow-up email whenever a new intake meeting is logged in Fireflies.ai.

Call Follow-Up Email Assistant

Today at 11:10 AM

Got it! I'll draft a follow-up email whenever a new intake call is transcribed and save it to your Gmail drafts, tailored to your agency's services and tone.

Call Follow-Up Email Assistant

 Agents

[Templates](#) > Agents

Agents templates



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 Support



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Support Email Agent

21h ago



Support Email Agent

21h ago

- + Create
- Home
- Automations
- Folders
- Favorites
- Templates
- App connections
- MCP servers

New
- Zap history
- More

Zaps

Trash

+ Create

Owner is Me

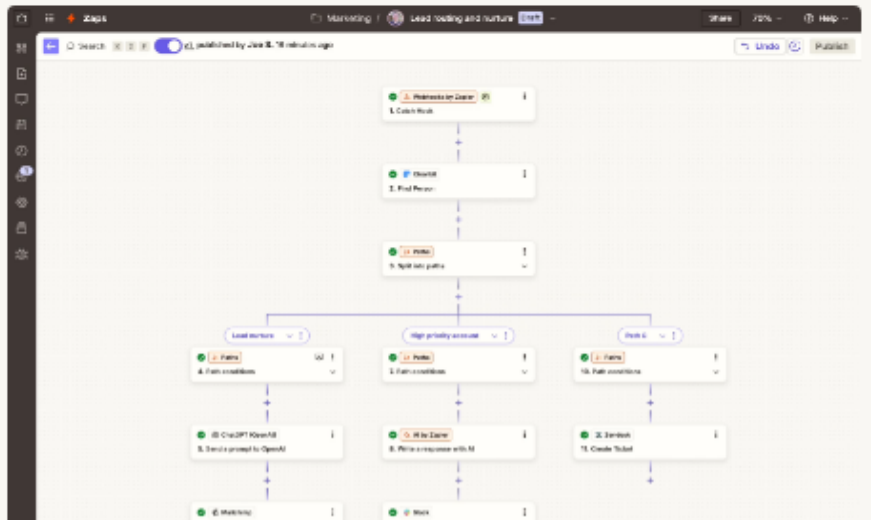
 Search by name, ID, or webhook

Filters

 Zaps

Start an automation by connecting two apps, a trigger and an action. Zapier integrates instantly with over 7,000 apps.

+ Create Zap



Search Ctrl ↑ F

Prompt

Copilot

Describe your workflow



Build

Start building ↑

Copilot is AI and can make mistakes. Please double-check responses.

or

Workflow

Trigger

1. Select the event that starts your Zap



Action

2. Select the event for your Zap to run

zapier

+ Create



Zaps
Automated workflows



Tables
Automated data storage

Search Ctrl ↑ F



Prompt

Copilot

Describe your workflow



Build

Start building ↑

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zapier

+ Create



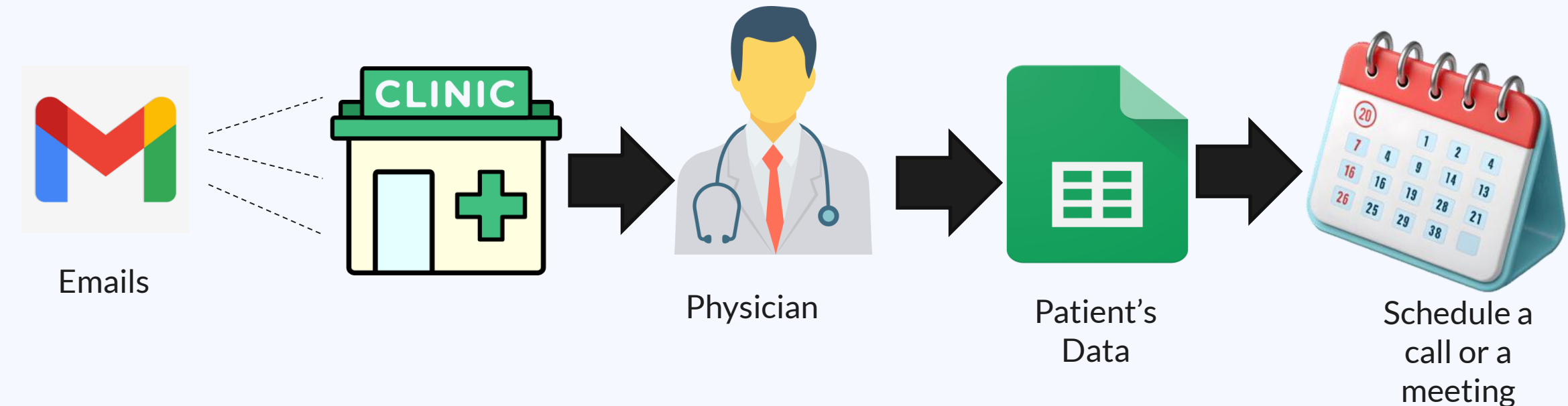
Zaps
Automated workflows



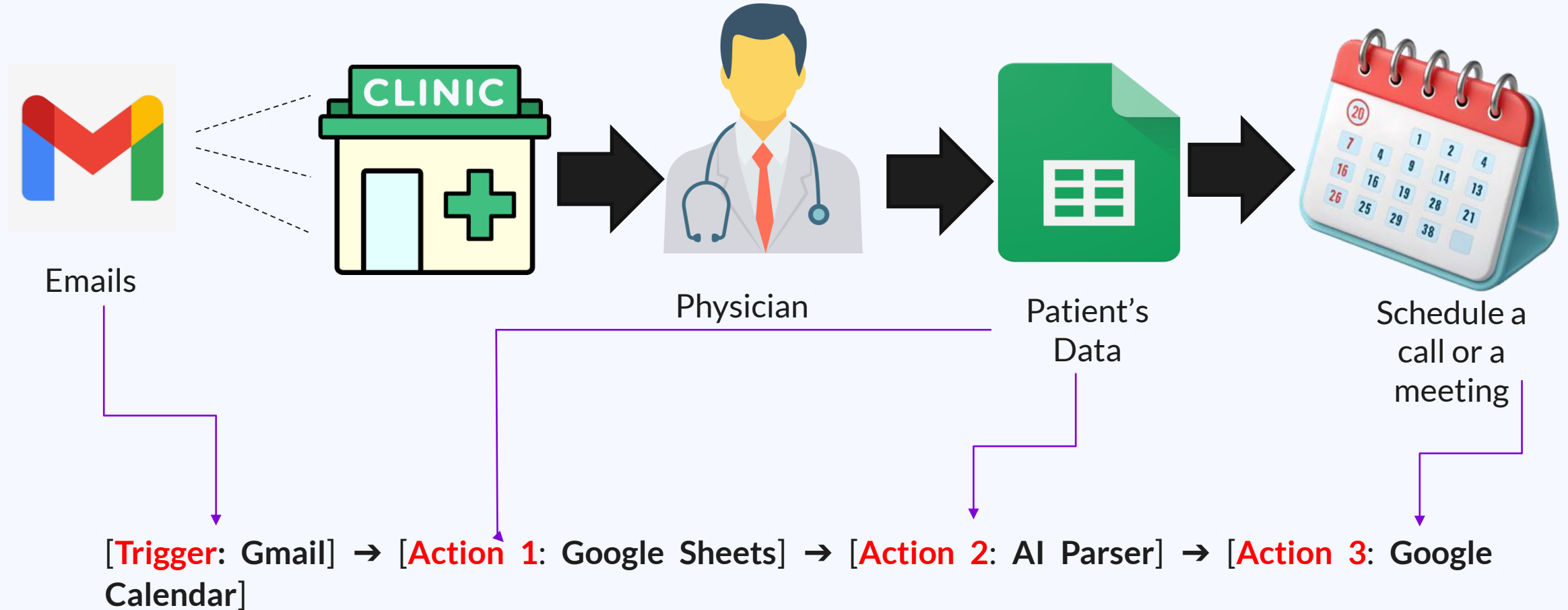
Tables
Automated data storage

Case Study

A outpatient clinic receives follow-up requests and lab result inquiries via email. The attending physician manually reads emails, logs patient IDs into a spreadsheet, schedules follow-up calls, and drafts responses. This creates administrative delay and risks missing urgent patient symptoms.



Case Study



Case Study

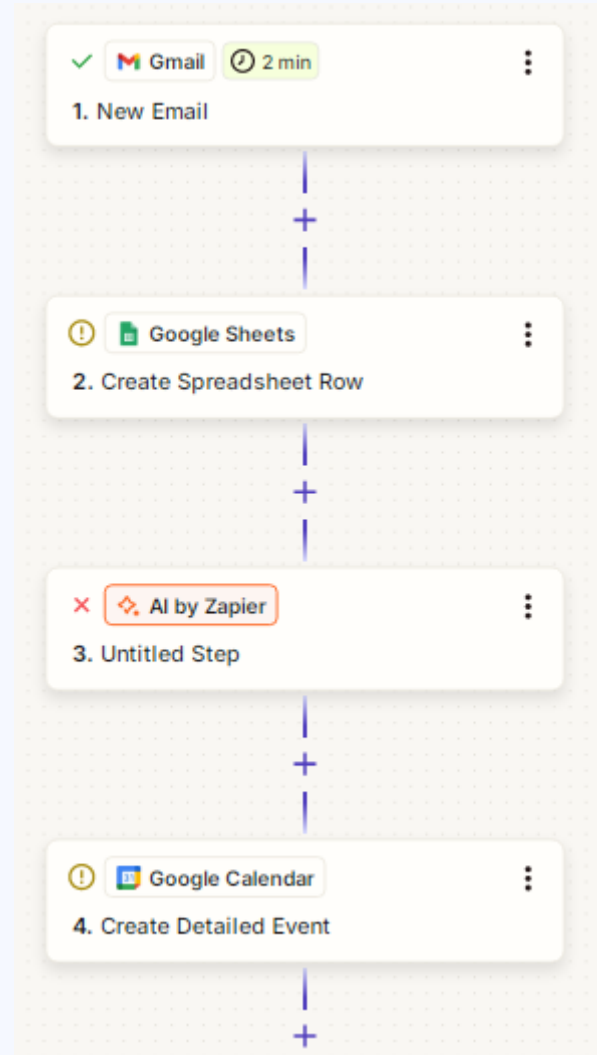
Using Zapier, builds a 4-step automated pipeline connecting Gmail, Google Sheets, and Google Calendar.

[Trigger: Gmail]

[Action 1: Google Sheets]

[Action 2: AI Parser]

[Action 3: Google Calendar]





9

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What would make your day easier, Sara?

Copilot

Enter an idea or app name to get started

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Start from scratch

- Zap**
Automated workflows
- Agent**
AI teammates
- Chatbot**
AI-powered chatbot
- MCP**
AI tool integrations
- Form**
Automation-ready forms
-

Case Study

Using Zapier, builds a 5-step automated pipeline connecting Gmail, Google Sheets, Google Calendar, and WhatsApp Notifications.

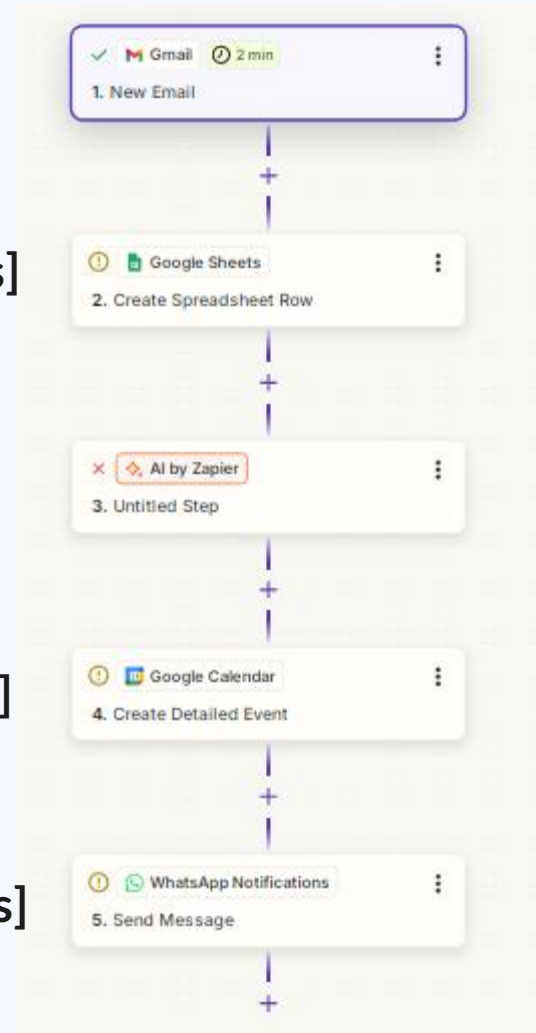
[Trigger: Gmail]

[Action 1: Google Sheets]

[Action 2: AI Parser]

[Action 3: Google Calendar]

[Action 4: WhatsApp Notifications]



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⚡ Zaps

Sara El-Metw...

SE

Clinic Follow-up Email Pipeline: Gmail to Sheets, AI Parser, and Calendar

Draft

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▶ Test run

Publish

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1. New Email

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⚠ Google Sheets

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2. Create Spreadsheet Row

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✗ AI by Zapier

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3. Untitled Step

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⚠ Google Calendar

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4. Create Detailed Event

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8 TRIAL DAYS LEFT

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What would you like to automate, Sara?

Copilot

Enter an idea or app name to get started

Microphone icon and Send icon

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Start from scratch

- Zap**
Automated workflows
- Agent**
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↶ Undo 🕒 ▶ Test run Publish

⚡ Copilot

Describe your workflow

🗣️ ▶ Build Start building ⬆

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or

⚡ Trigger

1. Select the event that starts your Zap

+

⚡ Action

2. Select the event for your Zap to run

+

📖 Library

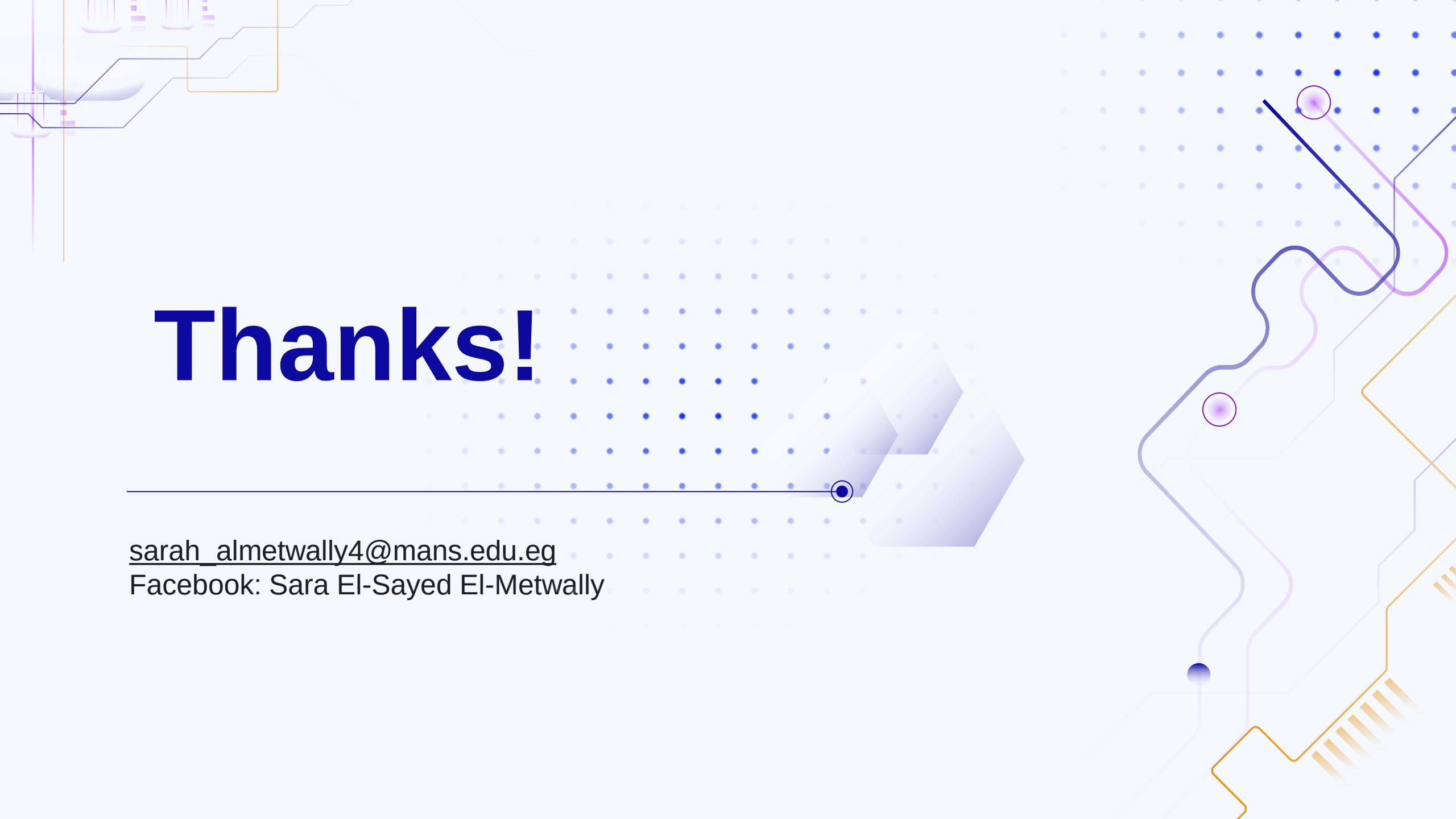
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🔍

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Thanks!

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